

Project Proposal: Swiftcore

Course: EC7207 High Performance Computing

Group Number: 50

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Project Title: Student Marks Analytics Engine (OpenMP + POSIX Threads)

Technologies: C (OpenMP, Pthreads), MongoDB, NextJS (frontend)

Project Overview

This project develops a **student marks management system** and a high-performance **analytics engine** to compute complex statistics from marks data and evaluate students. A NextJS frontend is used to add students, subjects, and marks, while a C backend stores data in a MongoDB database and runs analytics using **(1) Serial C**, **(2) OpenMP**, and **POSIX Threads** implementations. The main objective is to demonstrate how shared-memory parallel programming improves performance for computationally heavy statistical calculations, while maintaining accuracy close to the serial baseline.

Problem Description

A class can contain N students and M subjects. Marks can be represented as a matrix $\text{marks}[N][M]$ where each entry is a score (0–100), with possible missing values. When N and M increase (e.g., $N = 10,000$ students, $M = 100$ subjects), advanced analytics such as correlation matrices, bootstrapping, and ranking become computationally expensive. Therefore, we parallelize the analytics computations using OpenMP and Pthreads to reduce runtime compared to serial execution.

Statistical Calculations

- Descriptive statistics (per subject and overall)
 - Mean, variance, standard deviation
 - Min/Max, pass rate
 - Histogram
- Ranking
 - Z-score
 - Top K students and percentiles
- Subject–subject relationship
 - Covariance matrix and correlation matrix

Implementation Plan

- Set up Frontend to add data and display statistics and setup database.
- Do all calculation with serial code to compare with parallel code
- Do calculations parallelly with Open MP or Pthreads to get better performance than serial code.